



Laminar burning characteristics of ammonia and hydrogen blends at elevated initial pressures up to 2.5 MPa

Ning Wang^a, Tie Li^{a,b,*}, Xinpeng Guo^a, Zehao Wu^a, Shuai Huang^{a,b}, Xinyi Zhou^{a,b}, Shiyan Li^{a,b}, Run Chen^{a,b}

^a State Key Laboratory of Ocean Engineering, Shanghai Jiao Tong University, PR China

^b Institute of Power Plants and Automation, Shanghai Jiao Tong University, PR China

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ABSTRACT

Ammonia is a highly promising alternative fuel, and blending it with hydrogen can mitigate its poor combustion performance. However, the combustion mechanism of ammonia-hydrogen mixtures requires further validation through measurements of laminar burning velocity, particularly under high-pressure conditions relevant to practical combustors. Experimental data at high initial pressures (>1.0 MPa) are notably scarce. In this study, experiments were conducted using an outwardly propagating spherical flame in a high-pressure, high-temperature, constant-volume combustion chamber to determine the laminar burning velocities of ammonia-hydrogen blends at elevated initial pressures. The effects of different equivalence ratios ($0.6 \sim 1.4$), volumetric hydrogen ratios ($0.1 \sim 0.6$), and initial pressures ($0.1 \sim 2.5$ MPa) were evaluated. Chemical kinetics of ammonia and hydrogen combustion was investigated at high pressures. The experimental results were compared with the simulation results under a wide range of conditions using several existing mechanisms to evaluate their applicability. The results show that the laminar burning velocity increases first and then decreases as the equivalence ratio (ϕ) increases and reaches a maximum value around ϕ of 1.1. The laminar burning velocity increases non-linearly with the volumetric hydrogen ratio, with more pronounced acceleration at higher hydrogen ratios. In contrast, the laminar burning velocity decreases non-linearly with the increasing initial pressure, and this reduction becomes progressively slower as the initial pressure rises. The predictive performance of the mechanisms by Okafor et al. and Gotama et al. is relatively satisfactory under certain conditions, but further optimization based on experimental data is necessary. Additionally, reaction pathway analysis of ammonia-hydrogen combustion indicates that introducing hydrogen increases the concentration of active radicals such as OH, O, and H, thereby enhancing ammonia combustion. Sensitivity analysis of the laminar burning velocity identified the critical elementary reactions across varying pressures, providing strong support for optimizing ammonia-hydrogen chemical kinetics models and developing simulation models for practical combustion systems.

1. Introduction

The impact of greenhouse gases, such as carbon dioxide, on the environment is becoming increasingly severe, making the reduction of greenhouse gas emissions a critical goal for countries and industries worldwide. The adoption of alternative fuels has emerged as a pivotal green solution for the transportation sector. Among various alternatives, ammonia stands out due to its property to be burned without producing carbon dioxide and its favorable storage and transport characteristics, making it one of the most promising options [1,2]. However, challenges such as low chemical reactivity, slow flame propagation, and a narrow

flammability range can lead to reduced heat release rates and potential combustion instability, resulting in unburned ammonia emissions [3]. Consequently, to ensure the efficient and clean operation of ammonia-fueled combustion systems, it is imperative to investigate and develop appropriate combustion strategies.

1.1. Literature review

Ammonia, in conjunction with other highly reactive fuels such as diesel, gasoline, methane, and hydrogen, can enhance combustion performance [4,5]. Notably, hydrogen is a green fuel that does not emit carbon dioxide during combustion. Its high reactivity allows for

* Corresponding author at: 800 Dongchuan Rd., Shanghai 200240, PR China.
E-mail address: litie@sjtu.edu.cn (T. Li).

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Nomenclature		$S_{L,0}$	laminar burning velocity at the reference initial pressure state [cm/s]
A	projection area of the flame	T_{ad}	adiabatic flame temperature [K]
C_p	specific heat at constant pressure [W/kg/K]	T_0	initial ambient temperature [K]
CVCC	constant volume combustion chamber	V_a	volume of ammonia in the mixed fuel [m ³]
L_b	Markstein length [cm]	V_C	volume of the CVCC [L]
Le_a	Lewis number of ammonia [-]	V_h	volume of hydrogen in the mixed fuel [m ³]
Le_{eff}	effective Lewis number [-]	X_a	volumetric percentage of ammonia in the mixed fuel [%]
Le_h	Lewis number of hydrogen [-]	X_h	volumetric percentage of hydrogen in the mixed fuel [%]
m	number of initial pressure state points [-]	α	thermal diffusivity [10 ⁻⁵ m ² /s]
p	real-time ambient pressure in the CVCC [MPa]	β_p	pressure dependent coefficient [-]
P_0	initial ambient pressure [MPa]	$\Delta\beta_p$	uncertainty of the pressure dependent coefficient [-]
$P_{u,0}$	the reference initial pressure [MPa]	δ	flame thickness [10 ⁻⁴ m]
Q	apparent heat released during the combustion [J]	ε	stretch rate of the flame [1/s]
R	gas constant [-]	ϕ	equivalence ratio [-]
r_f	equivalent radius (r_f) of the flame [mm]	κ	isentropic index [-]
S	unstretched speed [cm/s]	λ	thermal conductivity [10 ⁻² W/m/K]
S_b	stretched speed [cm/s]	ρ_b	density of burned gas [kg/m ³]
S_L	laminar burning velocity [cm/s]	ρ_u	density of unburned gas [kg/m ³]
$\Delta S_{L,i}$	uncertainty of the laminar burning velocity		

effective mixing with ammonia to address the challenges associated with ammonia combustion and accelerate the overall combustion rate. Furthermore, ammonia can act as a carrier for hydrogen, as it can be decomposed into hydrogen by ammonia reforming technology [6],

greatly facilitating the utilization of ammonia-hydrogen mixtures. Recent studies have explored the application of these mixtures as fuels in practical combustion systems such as gas turbines and internal combustion engines. The utilization of ammonia-hydrogen mixtures has

Table 1

Previous studies on the laminar burning velocity of ammonia and hydrogen mixtures.

Year	Author	Reactants	P_0 (MPa)	T_0 (K)	ϕ (-)	X_h (-)	Method ^b
1987	Ronney et al. [12]	NH ₃ -air	0.006 ~ 0.2	298	0.7 ~ 1.5	—	SF
2000	Pfahl. et al. [13]	NH ₃ -air	0.1	295	0.5 ~ 1.1	—	SF
2004	Jabbour et al. [14]	NH ₃ -air	0.1	298	0.7 ~ 1.3	—	VC
2008	Takizawa et al. [15]	NH ₃ -air	0.105	298	0.9 ~ 1.2	—	SF
2015	Hayakawa et al. [16]	NH ₃ -air	0.1 ~ 0.5	298	0.8 ~ 1.2	—	SF
2018	Li et al. [17]	NH ₃ -air	0.05 ~ 0.3	298	0.8 ~ 1.3	—	SF
2020	Han et al. [18]	NH ₃ -air	0.1	298 ~ 448	0.85 ~ 1.25	—	HF
2024	Alvarez et al. [19]	NH ₃ -air	0.1 ~ 1.0	298	0.7 ~ 1.3	—	SF
2019	Mei et al. [20]	NH ₃ -N ₂ -O ₂	0.1 ~ 0.5	298	0.9 ~ 1.3	—	SF
2021	Shrestha et al. [21]	NH ₃ -N ₂ -O ₂	0.1 ~ 1.0	298 ~ 473	0.8 ~ 1.4	—	SF
2010	Lee et al. [22]	NH ₃ -H ₂ -air	0.1	298	0.6 ~ 1.67	0 ~ 0.5	SF
2010	Lee et al. [23]	NH ₃ -H ₂ -air	0.1	298	0.6 ~ 1.67	0.7 ~ 1	SF
2013	Kumar et al. [24]	NH ₃ -H ₂ -air	0.1	298	0.6 ~ 1.1	0.2~1.0	BCF
2014	Li et al. [25]	NH ₃ -H ₂ -air	0.1	298	0.6 ~ 1.4	0.23 ~ 0.6	BCF
2015	Ichikawa et al. [26]	NH ₃ -H ₂ -air	0.1 ~ 0.5	298	1	0.1 ~ 1	SF
2019	Han et al. [27]	NH ₃ -H ₂ -air	0.1	298	0.7 ~ 1.6	0 ~ 0.45	HF
2020	Lhuillier et al. [28]	NH ₃ -H ₂ -air	0.1	298 ~ 473	0.8 ~ 1.4	0 ~ 0.6	SF
2020	Wang et al. [29]	NH ₃ -H ₂ -air	0.1 ~ 0.5	298	0.6 ~ 1.6	0.4 ~ 0.6	HF
2020	Lesmana et al. [30]	NH ₃ -H ₂ -N ₂ -air	0.1	295	0.9 ~ 1.2	0 ~ 0.375 (0 ~ 0.5) ^a	VTF& BCF
2021	Wang et al. [31]	NH ₃ -H ₂ -air	0.1 ~ 0.5	360	0.7 ~ 1.2	0 ~ 0.2	SF
2021	Mei et al. [32]	NH ₃ -H ₂ -N ₂ -air	0.1 ~ 1.0	295	0.7 ~ 1.2	0 ~ 0.6 (0 ~ 0.8) ^a	SF
2022	Li et al. [33]	NH ₃ -H ₂ -air	0.05 ~ 0.2	298	0.8 ~ 1.4	0.2 ~ 1.0	SF
2022	Gotama et al. [34]	NH ₃ -H ₂ -air	0.1 ~ 0.5	298	0.8 ~ 1.8	0.4	SF
2022	Ji et al. [35]	NH ₃ -H ₂ -N ₂ -air	0.1	303	0.6 ~ 2.0	0 ~ 0.75 (0 ~ 1.0) ^a	SF
2022	Han et al. [36]	NH ₃ -H ₂ -N ₂ -air	0.1	298	0.6 ~ 1.6	0.02 ~ 0.34 (0.025 ~ 0.5) ^a	HF
2022	Zhang et al. [37]	NH ₃ -H ₂ -He-O ₂ -N ₂	0.1 ~ 1.0	298	0.7 ~ 1.5	0.5	SF
2022	Jin et al. [38]	NH ₃ -H ₂ -air	0.1 ~ 0.7	298 ~ 443	0.8 ~ 1.3	0.1 ~ 0.5	SF
2023	Chen et al. [39]	NH ₃ -H ₂ -air	0.05 ~ 0.15	298	0.5 ~ 1.5	0 ~ 1	SF
2023	Wu et al. [40]	NH ₃ -H ₂ -air	0.1	298	0.6 ~ 1.0	0.2 ~ 0.5	FF
2023	Yovino et al. [41]	NH ₃ -H ₂ -air	0.1 ~ 0.5	253 ~ 333	0.7 ~ 1.4	0.3,0.5	SF
2023	Wang et al. [42]	NH ₃ -H ₂ -O ₂ -air	0.1	303	0.6 ~ 1.8	0.2 ~ 0.5	SF
2023	S.Zitouni et al. [43]	NH ₃ -H ₂ -air	0.1	298	0.6 ~ 1.4	0 ~ 0.8	SF
2024	Shawnam et al. [44]	NH ₃ -H ₂ -air	0.1	300 ~ 750	0.9,1.0,1.1	0.3,0.4	SF

^a Dissociation ratio, NH₃ = 0.5 N₂ + 1.5H₂.

^b SF, Spherical flame, by constant volume combustion chamber; HF, Heat Flux; BCF, Bunsen Conical flame, by Bunsen burner; FF, Flat Flame, by McKenna burner; VTF, Vertical Tube Flame.

demonstrated stable combustion in spark-ignition (SI) internal combustion engines and gas turbines. Lhuillier et al. [7], Frigo et al. [8], and Dinesh et al. [9] collectively concluded that SI engines fueled by ammonia-hydrogen mixtures exhibit higher unburned ammonia emissions and elevated nitrogen oxide emissions compared to their gasoline counterparts. Valera-Medina et al. [10] investigated premixed ammonia/hydrogen swirl combustion for gas turbine applications, revealing that the addition of 30 % hydrogen enables stable combustion operation, although the overall combustion efficiency remains relatively low. Mashruk et al. [11] examined both swirling turbulent flames and laminar flames using a premixed ammonia/hydrogen/air mixture in a swirl burner, finding that NO and NO₂ emissions peaked at an equivalence ratio of approximately 0.8 before significantly decreasing on the rich side of the spectrum. Although ammonia-hydrogen blended fuel has demonstrated stable operation in certain combustion systems, the aforementioned research has revealed several persisting challenges, such as lower power output compared to traditional fuel engines, high levels of ammonia emissions, and excessive N₂O and NO emissions. To address these issues, it is crucial to understand the underlying mechanisms of ammonia combustion. In practical combustion systems such as gas turbines, internal combustion engines and boilers, the pressure range for ammonia-hydrogen combustion spans from slightly above atmospheric pressure to significantly elevated pressures, potentially reaching dozens of megapascals in high-power density engines. Consequently, extensive fundamental research is essential to investigate the combustion characteristics of ammonia and hydrogen, covering a wide pressure range, in order to optimize the operational modes of these devices and address existing challenges.

The laminar burning velocity, a fundamental characteristic of fuel combustion, serves as a critical parameter for calibrating chemical kinetics models and has been the subject of various studies. Table 1 provides a summary and organization of relevant researches on the laminar burning velocity of ammonia and ammonia-hydrogen mixtures. Since the 1980 s, Ronney et al. [12] have conducted the experimental studies on the laminar burning velocity of ammonia under various conditions. In the early 21st century, more efforts have been dedicated to the fundamental studies on ammonia combustion. Jabbour et al. [14] measured the laminar burning velocity of ammonia at atmospheric pressure and ambient temperature using the Vertical Tube Flame method. More recently, Li et al. [17] expanded the initial pressure range for measuring ammonia's laminar burning velocity to 0.3 MPa, while Hayakawa et al. [16] further increased it to 0.5 MPa. However, due to challenges in igniting ammonia under both lean and rich conditions, their equivalence ratio was limited to a range of 0.8 to 1.3. In recent studies, Alvarez et al. [19] broadened the pressure range for measuring laminar burning velocity in ammonia-air mixtures from 0.1 to 1.0 MPa. Lee et al. [22,23], investigating the acceleration effect of hydrogen on ammonia combustion, obtained the laminar burning velocities of ammonia-hydrogen mixtures at ambient conditions with hydrogen volume fractions ranging from 0 to 1. Following this work, Ichikawa et al. [26] assessed the laminar burning velocity of these mixtures at a pressure of 0.5 MPa, and more recently, Jin et al. [38] elevated the initial pressure for ammonia-hydrogen mixtures to 0.7 MPa. Furthermore, Mei et al. [32] investigated the laminar burning velocity of NH₃-H₂-N₂-air mixtures, where the hydrogen-to-nitrogen ratio was maintained at three to one, indicative of the proportion of ammonia decomposition products. Their experiments were conducted at a pressure up to 1.0 MPa. In summary, the studies presented in Table 1 employed various methodologies, including Spherical Flame, Heat Flux, Bunsen Conical Flame, Flat Flame, and Vertical Tube Flame, to assess the laminar burning velocities of ammonia-hydrogen mixtures. These studies examined the influence of different initial pressures (0.1 ~ 1.0 MPa), initial temperatures (295 ~ 750 K), equivalence ratios (0.5 ~ 1.67), and hydrogen volume fractions (0 ~ 1). Collectively, these investigations provided valuable data on laminar burning velocities that significantly contributed to optimizing chemical kinetics models for

ammonia-hydrogen combustion and advancing the understanding of combustion mechanisms involving ammonia-hydrogen.

1.2. Research gaps

Although the published works have concluded the laminar burning characteristics are strongly dependent on the ambient pressure, due to the limitations of the maximum allowable pressure of experimental apparatuses, all studies so far on the laminar burning velocity of ammonia-hydrogen have been conducted in the relatively low pressure environments. The highest recorded measurement for the laminar burning velocity of ammonia-hydrogen has only reached an initial pressure of 1.0 MPa [19,21,32,37], whereas the actual operating pressures of many combustion devices, including gas turbines and internal combustion engines, are significantly higher than those used in these experiments. The kinetic models developed thus far have only been validated under low-pressure conditions, and their applicability at high initial pressures encountered in real combustion systems remains unverified. The chemical kinetics of ammonia-hydrogen combustion under high pressure remains unclear. Therefore, obtaining experimental data on the laminar burning velocity of ammonia-hydrogen at elevated initial pressures and advancing our understanding of its combustion mechanism is essential for establishing the simulation model that optimizes the operation modes of ammonia-hydrogen-fueled combustion devices.

1.3. Motivation of this work

To address these gaps, this study conducted an experimental investigation of the laminar burning velocity of ammonia-hydrogen mixtures under a wide range of conditions using the outwardly spreading spherical flame method in a high-pressure, high-temperature constant-volume combustion chamber. The results include measurements of the laminar burning velocity up to the highest initial pressure of 2.5 MPa. The effects of different equivalence ratios (0.6 ~ 1.4), volumetric hydrogen ratios (0.1 ~ 0.6), and initial pressures (0.1 ~ 2.5 MPa) on the laminar burning velocity of the ammonia-hydrogen mixture were investigated. Additionally, this study compared the simulation results from the state-of-the-art kinetic mechanisms with the experimental results to assess the applicability of these mechanisms. The chemical reaction mechanisms of ammonia-hydrogen combustion at elevated pressures were investigated, leading to a clearer understanding of the associated chemical kinetics. Analysis of the reaction pathway of ammonia combustion and sensitivity of the elementary reactions was conducted to understand how hydrogen promotes ammonia combustion and to clarify the elementary reactions affecting the laminar burning velocity under different initial pressures. This analysis provides guidance for optimizing subsequent combustion mechanisms and developing numerical models for practical burners.

The structure of the paper is as follows. After the introduction, the paper describes the experimental setup, data processing methods, and simulation methods. In the Results and Discussion section, images of spherical flame diffusion under different conditions, as well as changes in pressure and heat release rate, are presented. This is followed by a discussion of the effects of different equivalence ratios, volumetric hydrogen ratios, and initial pressures, as well as an analysis of the reaction path of ammonia combustion and a sensitivity analysis of the elementary reactions. These analyses provide valuable data support for the development of subsequent combustion mechanisms and the establishment of numerical combustion models for actual burners. Finally, the paper concludes with a summary of the findings.

2. Experimental and numerical methods

2.1. Experimental methods

Fig. 1 shows the experimental apparatus used in this study. Details of the experimental apparatus have been discussed in our previous studies [31,45], and a brief description is provided here. The experimental investigation of laminar flame propagation was conducted in a high-temperature, high-pressure constant-volume combustion chamber (CVCC) with a designed pressure tolerance of 35 MPa. The internal volume of the CVCC is approximately 1.27 L and its shape is nearly a cube with a side length of 108 mm. Equipped with heating apparatus, pressure sensors, and temperature sensors for real-time monitoring of internal conditions, the CVCC provided an ideal environment for the experiment. In this study, two optical windows with a diameter of 100 mm were used within the CVCC. Additionally, two opposing spark plugs with a gap of 1 mm between them were employed. A stirring fan was strategically placed in one corner of the CVCC to ensure thorough gas mixing. During the experiment, the CVCC was initially heated to reach the desired target temperature, e.g., 423 K, which was the initial temperature of the mixing gas. Subsequently, ammonia, hydrogen, and air were sequentially introduced into the CVCC from high-pressure gas cylinders via the reducing valve and the needle valve. The mixing ratios and intake volumes of these gases were determined based on distinct initial conditions and partial pressure laws, respectively. After reaching the respective initial target pressures inside the CVCC, the stirring fan was activated to ensure thorough mixing of the gases before being turned off for 30 s. Following this interval, ignition occurred through discharge from the spark plugs, which ignited the premixed gas, resulting in the formation and gradual outward spread of the flame kernels. This study utilized a high-speed camera to capture the process of flame propagation. Fig. 1 illustrates the optical path of the Schlieren method. Initially, the light emitted by the LED passes through a pinhole to create a point source, then travels through a convex lens to be parallel light through the CVCC, and is focused by another convex lens before entering the high-speed camera, enabling the recording of the entire flame propagation process.

The fuels of ammonia and hydrogen mixtures with varied proportions of hydrogen contents were used in the experiment. The volumetric hydrogen ratio (X_h) is defined as

$$X_h = \frac{V_h}{V_h + V_a} \quad (1)$$

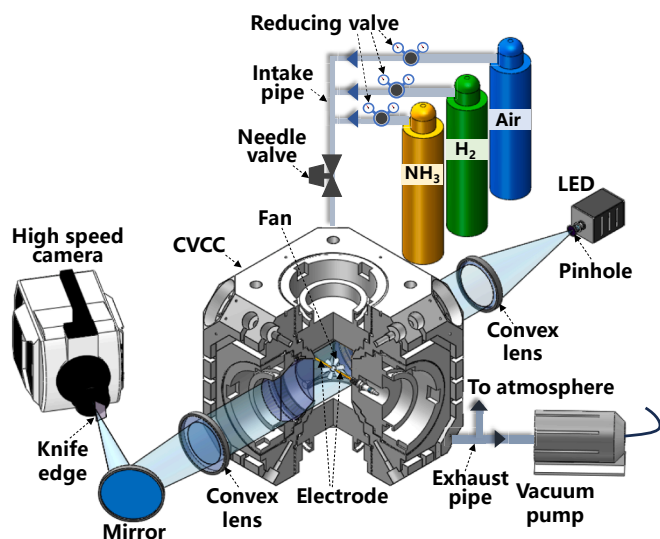


Fig. 1. Experimental apparatus.

where V_a and V_h are the volume fractions of ammonia and hydrogen in the fuel blends, respectively.

The experimental conditions are summarized in Table 2, encompassing investigations into different equivalence ratios (0.6 ~ 1.4), initial pressures (0.1 ~ 2.5 MPa), and volumetric hydrogen ratios (0.1 ~ 0.6). The initial temperature of the gas mixture was 423 K, and the ignition energy was 150 mJ. The images were recorded at a speed of 20,000 frames per second (fps) with a resolution of 512×512 pixels. The exposure time was set to 1.1 μ s.

2.2. Data processing methods

For the outwardly propagating spherical flame, both the linear extrapolation [31,46] and non-linear extrapolation [47] methods were used in this study. First, the projection area of the flame (A), obtained from the spherical flame images captured by the high-speed camera, is statistically analyzed to determine the equivalent radius (r_f). The threshold for processing flame images in this study is determined as 3 % of the maximum intensity value in the images after the background has been subtracted. The stretched flame speed (S_b) is defined as the rate of change of the flame radius with time,

$$S_b = \frac{dr_f}{dt} \quad (2)$$

Then, the stretching rate (ϵ) is obtained through,

$$\epsilon = \frac{d \ln A_f}{dt} = \frac{2}{r_f} \cdot \frac{dr_f}{dt} \quad (3)$$

Fig. 2 shows the results of the stretched flame speed versus the stretched rate of stoichiometric flame under an X_h of 0.3 and P_0 of 1.5 MPa. The symbols in Fig. 2 represent the experimental results obtained using Eqs. (2) and (3). As the flame gradually spreads, the stretched rate usually decreases over time. It can be observed that after ignition, the stretched flame speed decreases rapidly with the decrease in the stretched rate, followed by a stable propagation phase in the quasi-steady state when the flame has developed for a certain period, for example, when r_f is 4.5 mm in Fig. 2. After the quasi-steady regime, for instance, when the flame develops to an r_f of 18.5 mm in Fig. 2, the stretched flame speed changes rapidly with the stretching rate due to the influence of cellular instability. Therefore, to avoid the effects of ignition at the initial stage, and cellular instability and buoyancy effects at the later stage of flame propagation, this study uses the experimental results obtained during the quasi-steady state and selects data where the equivalent radius is between 8 and 15 mm. The data of laminar burning velocities in this study were all acquired after the ignition effect and before the occurrence of cell instability, thereby avoiding their influences.

For linear extrapolation, it is generally believed that there is a linear relationship between the stretched flame speed and the stretched rate,

$$S_b = S_b^0 - L_b \cdot \epsilon \quad (4)$$

where S is the unstretched flame speed and L_b is the Markstein length. For non-linear extrapolation, there is a non-linear relationship between

Table 2

Experimental conditions and imaging parameter settings.

Description (units)	Value
Experimental conditions	
Volumetric hydrogen ratio, X_h (—)	0.1 ~ 0.6
Equivalence ratio, ϕ (—)	0.6 ~ 1.4
Initial pressure, P_0 (MPa)	0.1 ~ 2.5
Initial temperature, T_0 (K)	423
Imaging parameter settings	
Schlieren imaging speed (fps)	20,000
Exposure time (μ s)	1.1
Image resolution (—)	512×512

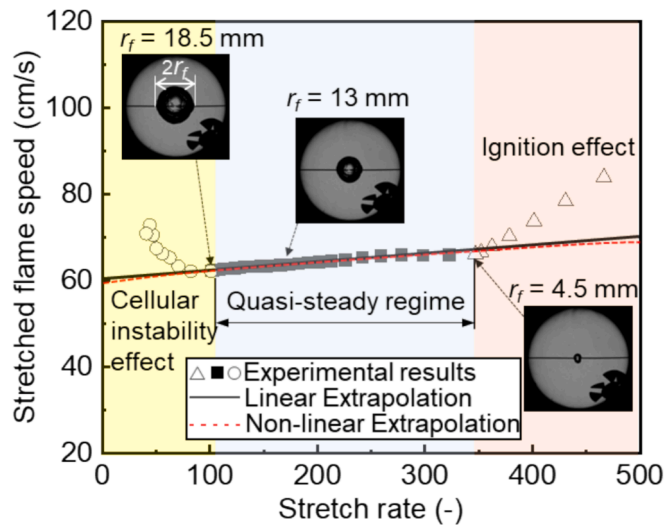


Fig. 2. The evolution of the stretched flame speed versus the stretch rate. The symbols indicate the experimental results and the lines are the extrapolation results. ($\phi = 1.0$; $X_{H_2} = 0.3$; $P_0 = 1.5$ MPa; $T_0 = 423$ K).

the stretched flame speed and the stretched rate,

$$\left(\frac{S_b^0}{S_b}\right)^2 \ln\left(\frac{S_b^0}{S_b}\right) = -2\left(\frac{L_b \cdot \varepsilon}{S_b}\right) \quad (5)$$

By fitting the experimental data according to the linear relationship or non-linear relationship and extrapolating to the stretched rate of 0, the non-stretched flame speed can be obtained, which is the intersection point of the fitting line and the vertical axis in Fig. 2. As shown, the difference between the linear extrapolation results and the non-linear extrapolation results is small in this study. Thus, only the linear extrapolation results are presented in the following sections.

From the non-stretched flame speed, the laminar burning velocity (S_L) can be further calculated,

$$S_L = S_b^0 \cdot \frac{\rho_b}{\rho_u} \quad (6)$$

where ρ_u is the density of unburned gas and ρ_b is the density of burned gas. In this study, the densities of unburned and burned gases are calculated using the kinetic model by Okafor et al. [51] in Chemkin-Pro software.

2.3. Numerical methods

The numerical study was conducted using Chemkin-Pro software, simulating laminar combustion in a one-dimensional flame using the flame speed module. The simulation conditions were set to match the experimental conditions. This study utilized the existing mechanisms developed in recent years by Shrestha et al. (with 125 species and 1099 reactions, validated at 298 ~ 473 K and 0.1 ~ 1.0 MPa) [21], Mei et al. (with 40 species and 257 reactions, validated at 298 K and 0.1 ~ 1 MPa) [32], Gotama et al. (with 32 species and 165 reactions, validated at 298 K and 0.1 ~ 0.5 MPa) [34], Song et al. (with 34 species and 204 reactions, validated at 450 ~ 900 K and 10 MPa by the mole fraction of different species in a laminar flow reactor) [48], Glaborg et al. (with 151 species and 1397 reactions, validated at 900 ~ 1500 K and 0.1 MPa) [49], Otomo et al. (with 32 species and 213 reactions, validated at 298 K and 0.1 ~ 0.5 MPa) [50], Okafor et al. (with 42 species and 130 reactions, validated at 298 K and 0.1 ~ 0.5 MPa) [51], Han et al. (with 35 species and 177 reactions, validated at 298 K and 0.1 MPa) [52], Stagni et al. (with 31 species and 203 reactions, validated at 500 K ~ 2000 K and 0.1 MPa) [53], Wang et al. (with 91 species and 444 reactions, validated at 298 K ~ 448 K and 0.1 MPa) [54], Zhang et al. (with 38

species and 263 reactions, validated at 303 K and 0.1 MPa) [55], and Zhou et al. (with 34 species and 204 reactions, validated at the internal combustion engine conditions by the emissions) [56] to obtain results such as laminar burning velocity, Lewis number, and flame thickness. Additionally, the reaction pathway analysis and sensitivity analysis of laminar burning velocity were conducted using the same model in Chemkin-Pro software. In the simulation, the absolute error was limited to 10^{-11} and the relative error was restricted to 10^{-5} during the computation. Under the premise of ensuring the convergence of the calculation results, Fig. 3 takes the results calculated using the mechanism by Okafor et al. [51] as an example to demonstrate the variations of the laminar burning velocity with different grid numbers under different initial pressures. It can be observed that when the maximum grid number is higher than 100, the laminar burning velocity is not sensitive to the increase in the grid number. In this study, the maximum grid number of 300 was used for the calculation, and thus the obtained results are grid-independent.

3. Results and discussion

3.1. Flame morphology

Fig. 4 displays the evolution of the outwardly propagating spherical flame under the different initial pressures (0.5 ~ 2.5 MPa), volumetric hydrogen ratios (0.5 ~ 0.7), and equivalence ratios (0.8 ~ 1.4). It illustrates the progression of spherical flames from r_f of 5 mm to 25 mm. Fig. 5 shows the development of the projection area of the spherical flame under the same conditions as those in Fig. 4, and the rate of change of the flame's projection area indicates the speed of flame propagation. With an increase in the initial pressure, the flame propagation speed decreases. When the volumetric hydrogen ratio is increased from 0.3 to 0.5, the flame area changes noticeably faster. Additionally, the flame area increases more rapidly for the stoichiometric flame when the equivalence ratio is 1.0, compared to when it is 0.8 or 1.4 under lean or rich conditions, respectively.

It is evident from Fig. 4 that at the volumetric hydrogen ratio of 0.7, an increase in the initial pressure results in a deepening of the color of the burned region within the spherical flame. This is attributed to the larger density difference between the burned and unburned regions under the higher initial pressures, which leads to a more pronounced burned region in the Schlieren images. Furthermore, at initial pressures of 1.5 and 2.5 MPa, the flame surface is observed to be less smooth compared to when the initial pressure is 0.5 MPa. Notably, when the flame develops to an r_f of 20 mm under an initial pressure of 2.5 MPa, its surface begins to wrinkle due to the instability induced by high-pressure

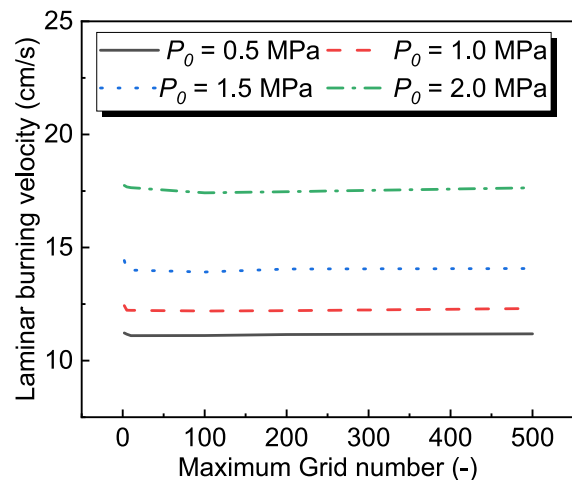


Fig. 3. Numerical results of the laminar burning velocities under different maximum grid numbers. ($X_{H_2} = 0.3$; $\phi = 1.1$).

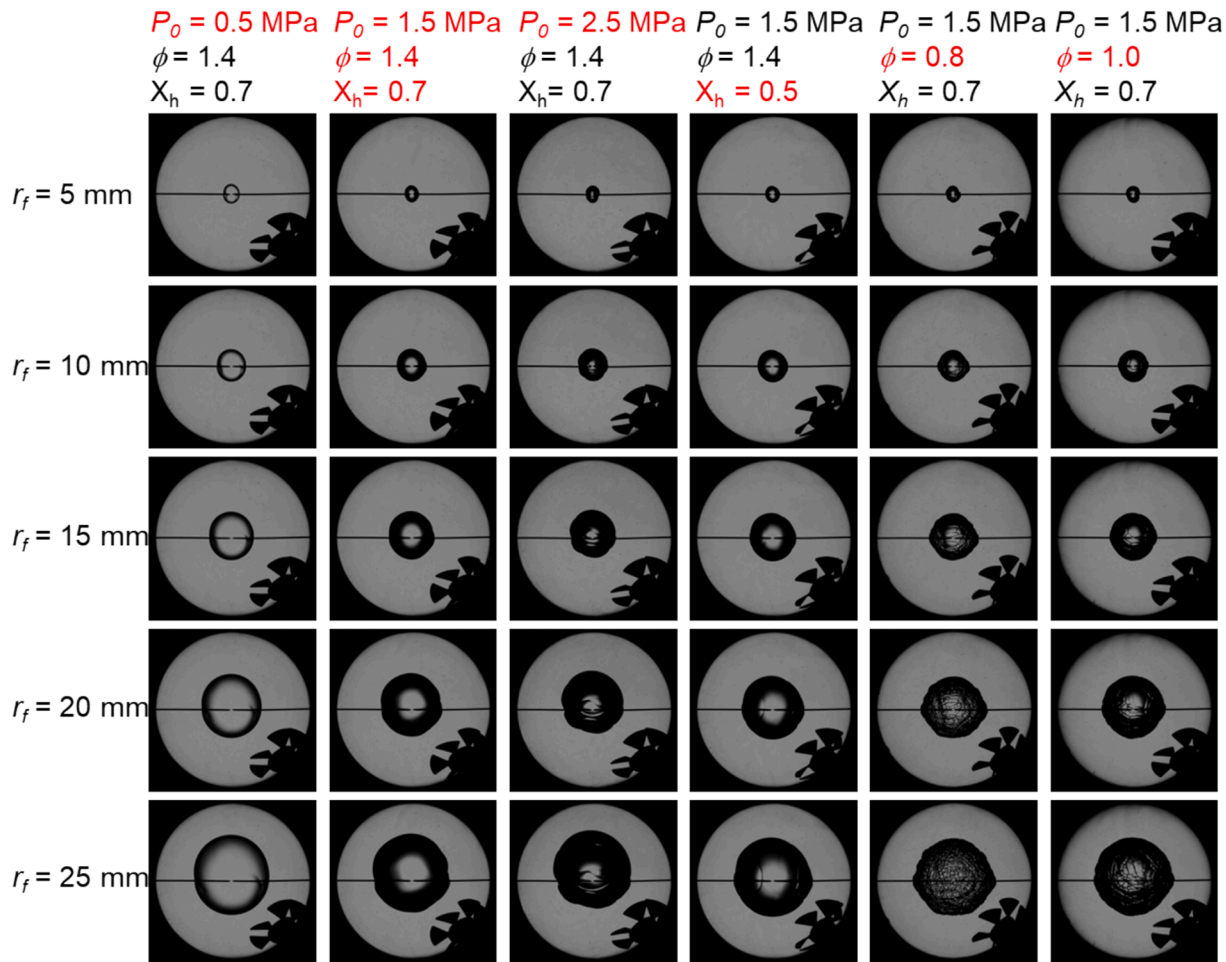


Fig. 4. Temporal development of outwardly propagating spherical flame of the $\text{NH}_3 + \text{H}_2$ blend by Schlieren. ($P_0 = 0.5, 1.5, 2.5$ MPa; $\phi = 0.8, 1.0, 1.4$; $X_h = 0.5, 0.7$; $T_0 = 423$ K).

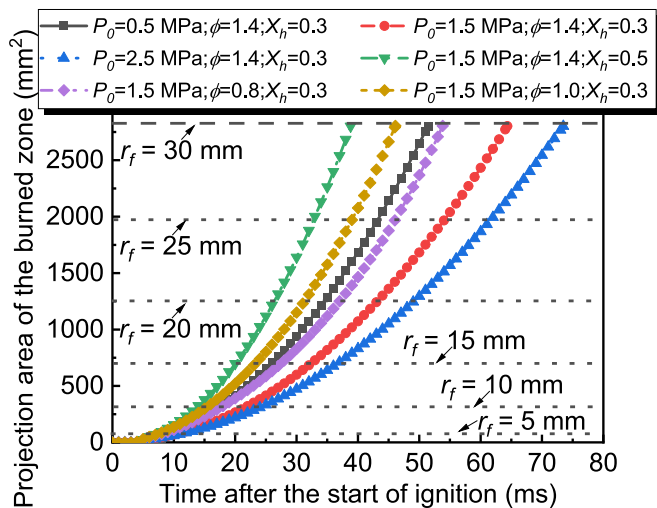


Fig. 5. Development of the projection area of the burned zone. ($P_0 = 0.5, 1.5, 2.5$ MPa; $\phi = 0.8, 1.0, 1.4$; $X_h = 0.5, 0.7$; $T_0 = 423$ K).

conditions. Additionally, the slower flame propagation under the high pressure and the smaller density of the burned region lead to a buoyancy effect. At an r_f of 15 mm under the initial pressures of 1.5 and 2.5 MPa, the center of the spherical flame rises above the location where the flame kernel formation occurs, i.e. the center of the opposing spark electrodes,

with this phenomenon becoming more pronounced as the flame propagates further. As the volumetric hydrogen ratio increases to 0.5, the buoyancy effect on the flame is reduced due to the faster flame propagation, while the flame becomes more susceptible to disturbances. As shown in the figure, when the spherical flame develops to an r_f of 20 mm, the flame surface begins to show unevenness at a volumetric hydrogen ratio of 0.5. As the equivalence ratio decreases, the flame surface wrinkles become more obvious. Specifically, at an equivalence ratio of 0.8, the flame surface begins to show slight wrinkles when the spherical flame develops to an r_f of 10 mm. As the flame propagates outward, these wrinkles become more pronounced, due to the flame being more susceptible to instability under lean conditions.

Fig. 6 depicts the development of pressure in the CVCC and the apparent heat release rate after the start of ignition. According to the first law of thermodynamics and the ideal gas law, the apparent heat release rate during combustion in the CVCC can be calculated using,

$$\frac{dQ}{dT} = \frac{1}{\kappa - 1} V_C \frac{dp}{dt} \quad (7)$$

where Q represents the apparent heat released during combustion; κ denotes the isentropic index, obtainable from the NIST website [57]; V_C is the volume of the CVCC, and p indicates the real-time pressure in the CVCC.

As depicted in Fig. 6, after ignition, the ammonia-hydrogen-air mixture starts combustion, leading to heat release and a gradual increase in pressure within the CVCC. Subsequently, once a state of equilibrium between the heat release from fuel combustion and

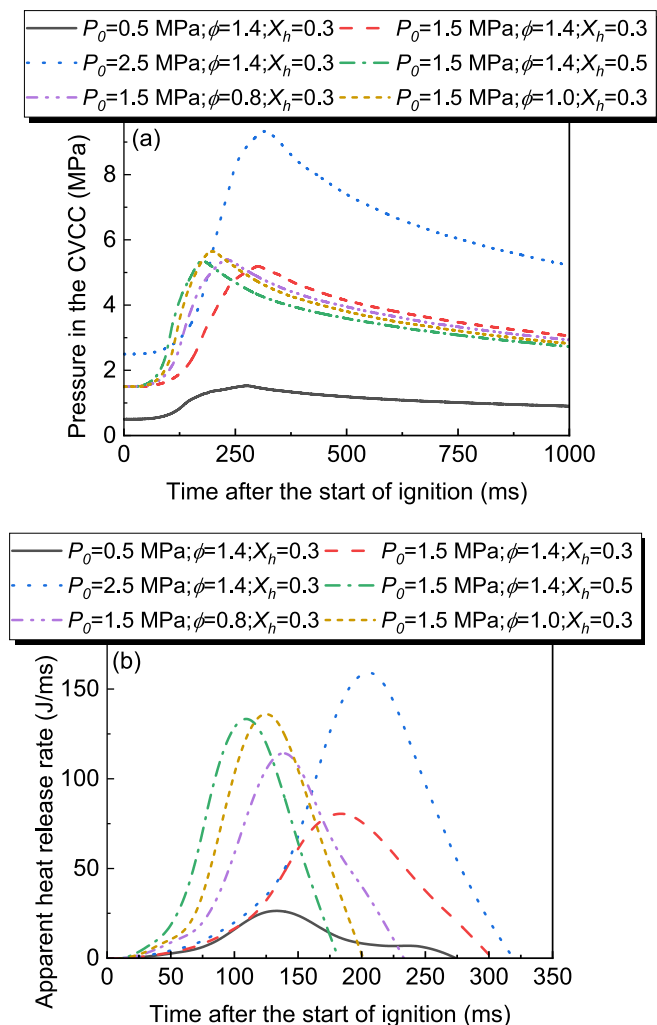


Fig. 6. Development of (a) the pressure in the CVCC and (b) the apparent heat release rate after the start of ignition. ($P_0 = 0.5, 1.5, 2.5$ MPa; $\phi = 0.8, 1.0, 1.4$; $X_{H_2} = 0.5, 0.7$; $T_0 = 423$ K).

environmental heat loss is achieved, the pressure in the CVCC reaches its peak before gradually declining as the heat generated by the fuel combustion becomes less than the heat lost to the surroundings. Under the conditions where the initial pressure is 1.5 MPa and the volumetric hydrogen ratio is 0.5, an accelerated rise in the pressure and an earlier attainment of the peak heat release rate are observed. This is due to the increased chemical reactivity of hydrogen, which accelerates the combustion process. Decreasing the volumetric hydrogen ratio to 0.3 results in a slower rate of pressure increase and a delayed peak heat release time. Moreover, during the initial stage (0 ~ 120 ms after ignition), when the initial pressures are increased from 0.5 MPa to 1.5 MPa, the development of heat release rates exhibits notable similarity across all scenarios, owing to the greater total fuel quantities despite the slower flame propagation speeds at the higher pressures, as indicated in Fig. 6. The larger amounts of released energy from the increased fuel quantities at the higher initial pressures result in the elevated peak values for both the heat release rates and pressures in the CVCC, with the peak pressure even reaching approximately 9.0 MPa at an initial pressure of 2.5 MPa. Furthermore, when considering a volumetric hydrogen ratio of 0.3 at an initial pressure of 1.5 MPa, it is noted that the pressure in the CVCC increases more rapidly for the stoichiometric combustion (with an equivalence ratio of 1.0) compared to lean or rich conditions (with equivalence ratios of 0.8 or 1.4), resulting in the faster rates of heat release. This can be attributed to the faster flame propagation and

reaction rates for the stoichiometric flame.

3.2. Effects of equivalence ratios

Figs. 7 and 8 show the laminar burning velocities of the ammonia and hydrogen blend versus equivalence ratios under various volumetric hydrogen ratios at the initial pressures of 1.5 MPa and 2.5 MPa, respectively. The symbols represent the experimental results, while the lines depict the numerical results based on different kinetic models.

As shown in Figs. 7 and 8, the laminar burning velocity increases first and then decreases as the equivalence ratio increases, reaching its maximum value around an equivalence ratio of 1.1. At the higher volumetric hydrogen ratios, the cellular instability tends to occur more readily in the lean conditions, becoming more pronounced as the initial pressure increases. This instability makes it challenging to obtain statistically reliable experimental values for the laminar burning velocity. Therefore, in cases with the higher volumetric hydrogen ratios, this study only presents results for the laminar burning velocity under the relatively rich conditions. As illustrated in Fig. 7, at an initial pressure of 1.5 MPa, when the volumetric hydrogen ratio is 0.1 and 0.2 in the lean conditions, i.e. when the equivalence ratio is less than or equal to 1, the calculated results from the various mechanisms, by Okafor et al. [51], Han et al. [52], Stagni et al. [53], Mei et al. [32], Wang et al. [54], Zhang et al. [55], etc., are relatively close to the experimental values, with particularly good agreement from the mechanisms of Mei et al. [32] and Wang et al. [54]. However, in the relatively rich conditions, i.e. when the equivalence ratio is greater than 1, all mechanisms used in this study predict the higher laminar burning velocities than the experimental results. When the volumetric hydrogen ratio is increased to 0.3 and 0.4 in these conditions, and the equivalence ratio is greater than or equal to 1.0, the predictions based on the mechanisms of Okafor et al. [51] and Gotama et al. [34] are relatively close to the experimental results, while the predictions from other mechanisms are higher. As the volumetric hydrogen ratio is further increased to 0.5 and 0.6, in relatively rich conditions, all predictions except those based on the mechanism of Okafor et al. [51] are higher than the experimental results.

When the initial pressure is 2.5 MPa, as illustrated in Fig. 8, the calculated results derived from the mechanisms of Han et al. [52], Stagni et al. [53], Mei et al. [32], Wang et al. [54], Zhang et al. [55], and Gotama et al. [34] are relatively close to the experimental values at the volumetric hydrogen ratios of 0.1 and 0.2 under the lean conditions when the equivalence ratio is less than or equal to 1.0. In the rich conditions where the equivalence ratio exceeds 1.0, the mechanisms proposed by Gotama et al. [34], Wang et al. [54], and Han et al. [52] demonstrate good predictive performance. As the volumetric hydrogen ratio increases to 0.3 and 0.4 within these rich conditions, the predicted results based on the mechanisms proposed by Okafor et al. [51], Han et al. [52], Wang et al. [54], and Gotama et al. [34] are relatively close to the experimental values. However, at the hydrogen volume fraction of 0.5, while the predictions from the mechanisms by Okafor et al. [51] and Gotama et al. [34] are slightly lower than the corresponding experimental values, all other mechanisms yield the predictions that exceed these experimental values.

Table 3 lists the experimental values of laminar burning velocities under different volumetric hydrogen ratios at the initial pressure of 1.5 MPa, as well as the relative errors between the experimental results (Exp.) and numerical results (Sim.) by the mechanisms of Okafor et al. [51], Gotama et al. [34] and Wang et al. [54]. The results show that the prediction of the three mechanisms are different under these conditions. The mechanisms by Gotama et al. [34] and Wang et al. [54] perform well at the volumetric hydrogen ratios of 0.1 and 0.2, while the mechanism by Okafor et al. [51] predicts the laminar burning velocities with an error of less than 4 % at the volumetric hydrogen ratios of 0.3 and 0.4.

In summary, the current mechanisms exhibit varying behaviors across the different equivalence ratios, volumetric hydrogen ratios, and initial pressures. At the low volumetric hydrogen ratios, particularly

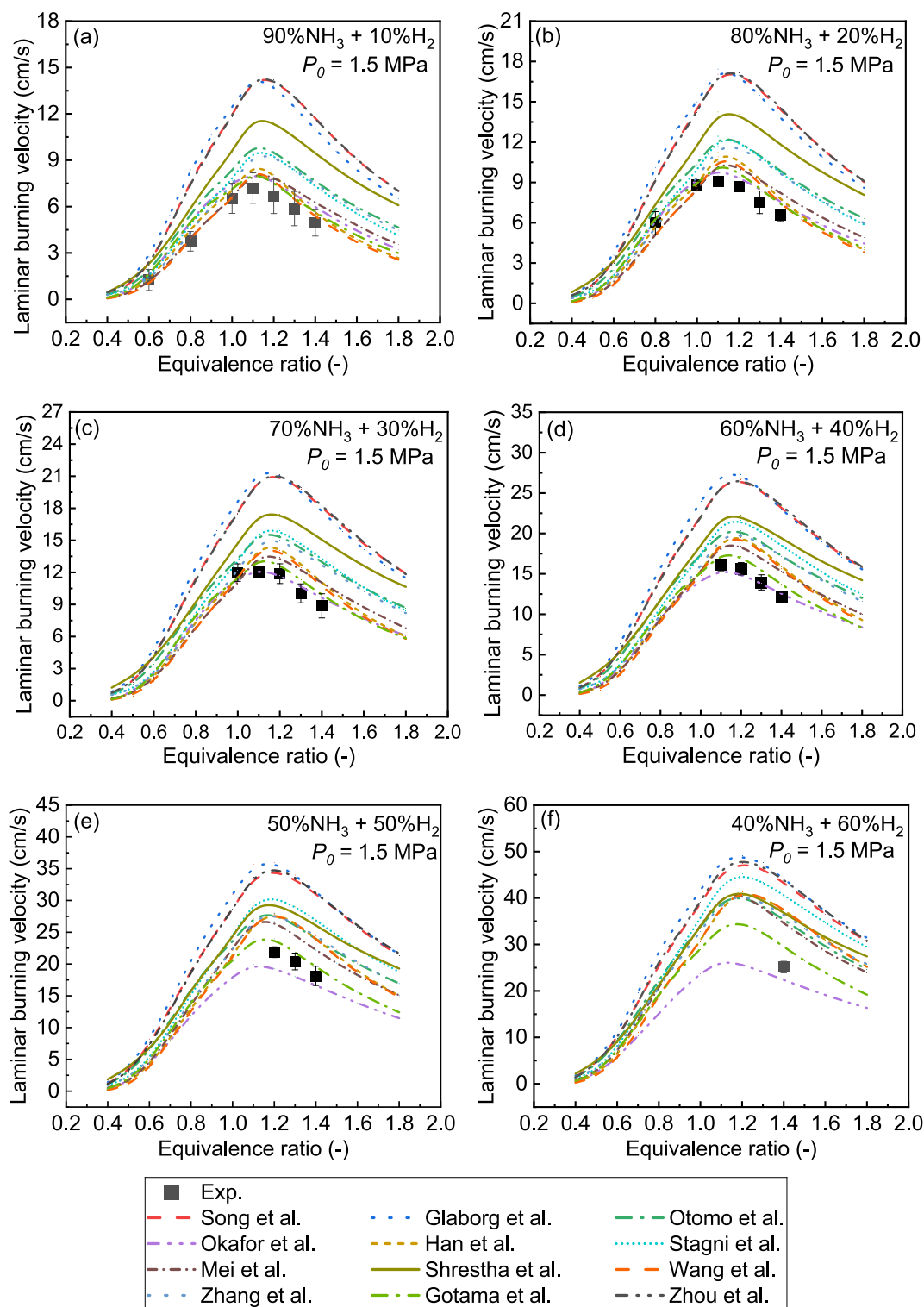


Fig. 7. Laminar burning velocities of (a) 90 % NH_3 + 10 % H_2 , (b) 80 % NH_3 + 20 % H_2 , (c) 70 % NH_3 + 30 % H_2 , (d) 60 % NH_3 + 40 % H_2 , (e) 50 % NH_3 + 50 % H_2 and (f) 40 % NH_3 + 60 % H_2 under different equivalence ratios at P_0 of 1.5 MPa. The symbols indicate the experimental results and the lines are the numerical results. ($P_0 = 1.5$ MPa; $T_0 = 423$ K).

under the lean conditions, the predicted values from several mechanisms closely align with the experimental results. However, as the volumetric hydrogen ratio increases to 0.3 and 0.4, there is a gradual decline in the number of mechanisms that yield accurate predictions. Notably, more effective predictive mechanisms are observed at an initial pressure of 2.5 MPa compared to 1.5 MPa. When the volumetric hydrogen ratio reaches 0.5 or higher, no suitable mechanism has been

identified that can accurately predict experimental outcomes. Overall, the mechanisms proposed by Okafor et al. [51] and Gotama et al. [34] demonstrate the superior performance at both the 1.5 MPa and 2.5 MPa initial pressures, especially at the volumetric hydrogen ratios of 0.3 and 0.4.

As mentioned earlier, the cellular instabilities can affect the acquisition of the laminar burning velocity. These instabilities are initially

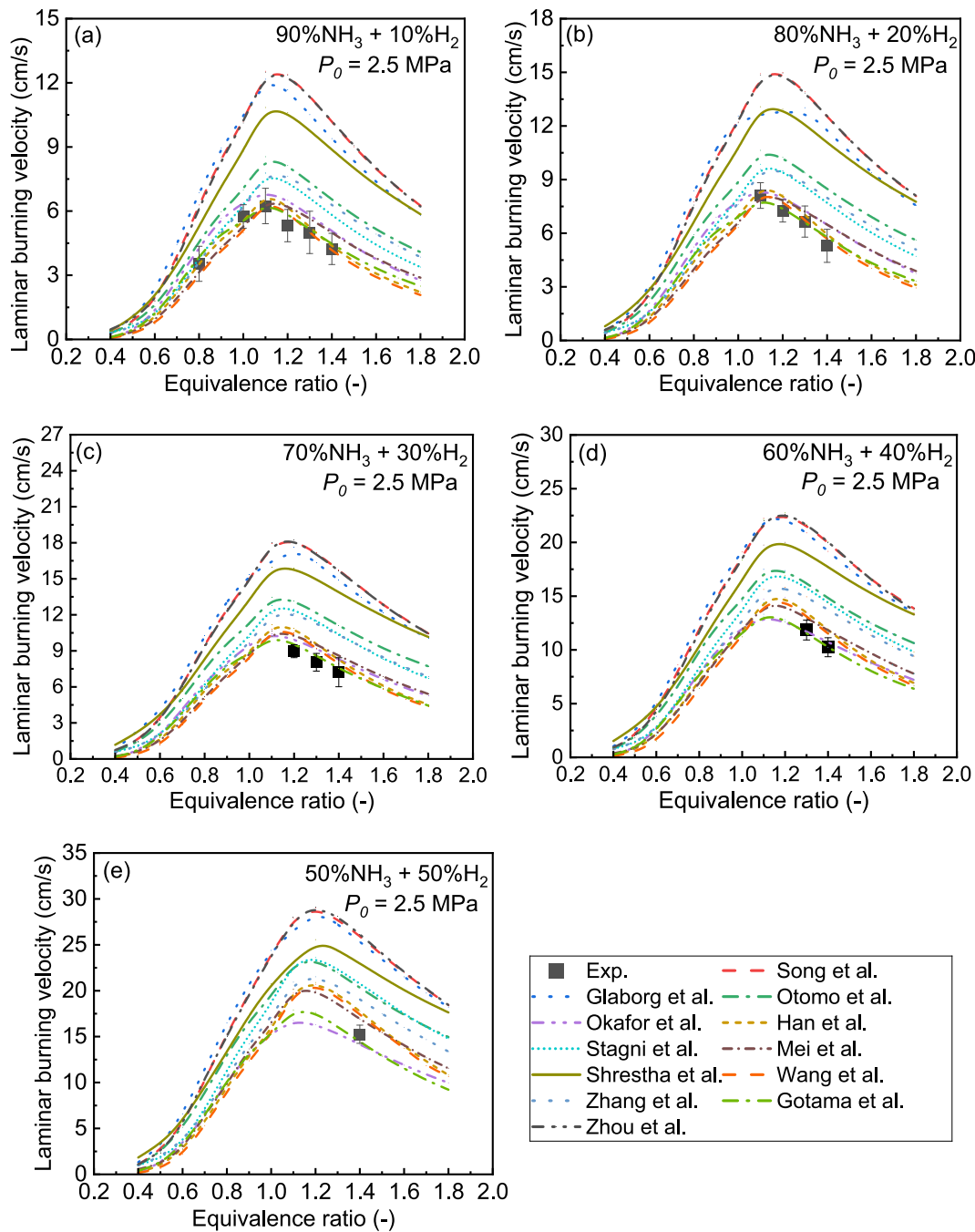


Fig. 8. Laminar burning velocities of (a) 90 % NH_3 + 10 % H_2 , (b) 80 % NH_3 + 20 % H_2 , (c) 70 % NH_3 + 30 % H_2 , (d) 60 % NH_3 + 40 % H_2 and (e) 50 % NH_3 + 50 % H_2 under different equivalence ratios at P_0 of 2.5 MPa. The symbols indicate the experimental results and the lines are the numerical results. ($P_0 = 2.5 \text{ MPa}$; $T_0 = 423 \text{ K}$).

Table 3

Comparison of experimental and numerical results of the laminar burning velocity. ($P_0 = 1.5 \text{ MPa}$, $\phi = 1.1$).

X_h (-)	Exp. (cm/s)	Relative errors between Sim. and Exp. (-)		
		Okafor et al.	Gotama et al.	Wang et al.
0.1	7.2	9.1 %	2.1 %	2.7 %
0.2	9.1	7.4 %	5.2 %	5.3 %
0.3	12.1	1.9 %	8.9 %	6.9 %
0.4	15.5	3.7 %	13.0 %	9.4 %

suppressed by the strong curvature of the flame but diminish or even extinguish as the flame expands. The incorporation of hydrogen exacerbates the prominence of cellular instability. To further investigate the

characteristics of flame instability, the effective Lewis number (Le_{eff}) and flame thickness (δ) under the different equivalence ratios and volumetric hydrogen ratios were obtained through the simulations based on the mechanism proposed by Okafor et al. [51], as shown in Figs. 9 and 10.

In this study, the flames are primarily influenced by two types of instability. The first arises from the disparity between the heat conduction and mass diffusion rates, resulting in the diffusional-thermal instability at the flame front. The second is the hydrodynamic instability, which is associated with the gas expansion during the exothermic reactions. Diffusional-thermal instability can be characterized by the effective Lewis number, where a lower Lewis number indicates increased susceptibility to this form of instability [58]. Hydrodynamic

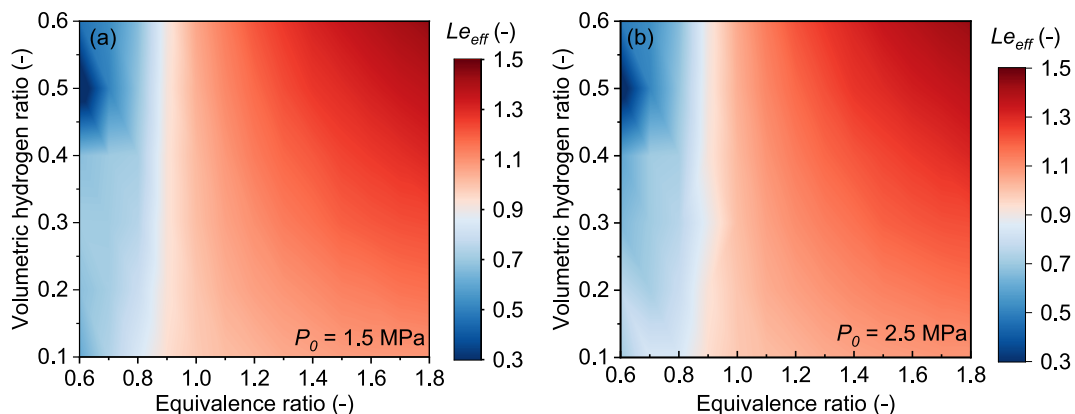


Fig. 9. The effective Lewis number under different equivalence ratios and volumetric hydrogen ratios at P_0 of (a) 1.5 MPa and (b) 2.5 MPa.

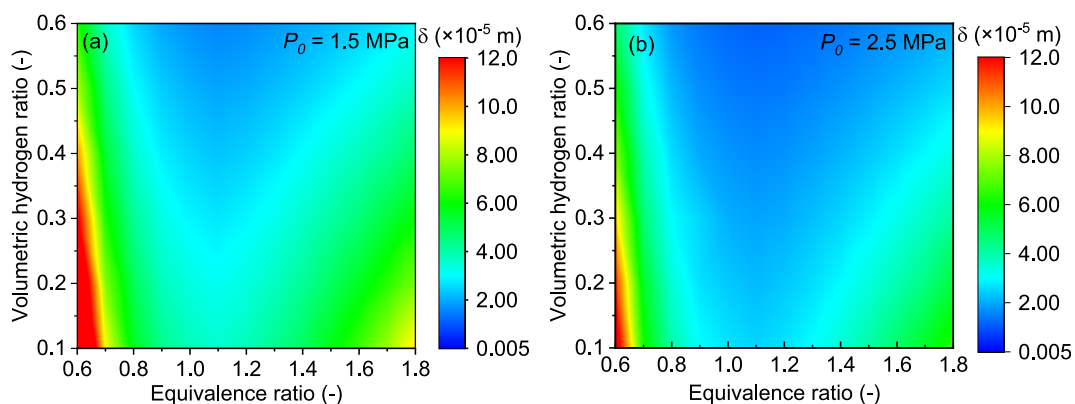


Fig. 10. The flame thickness under different equivalence ratios and volumetric hydrogen ratios at P_0 of (a) 1.5 MPa and (b) 2.5 MPa.

instability can be assessed through flame thickness, where a reduced flame thickness enhances vulnerability to wrinkles [58]. The comprehensive methodology for calculating the effective Lewis number is thoroughly detailed in our previous study [31], and a concise overview is provided here. For the ammonia and hydrogen mixtures, this paper employs the effective Lewis number for the multi-component fuels, calculated using a volume-weighted approach,

$$\frac{1}{Le_{eff}} = \frac{X_a}{Le_a} + \frac{X_h}{Le_h} \quad (8)$$

where X_a and X_h are the volume fractions of ammonia and hydrogen in the mixed fuel, and Le_a and Le_h are the Lewis numbers corresponding to ammonia and hydrogen, respectively, which can be obtained by the simulation in Chemkin-pro.

The flame thickness is the thermal diffusion flame thickness, which is typically calculated using the following formula,

$$\delta = \frac{\lambda}{\rho_u C_p S_L} = \frac{\alpha}{S_L} \quad (9)$$

where λ is the thermal conductivity; C_p is the specific heat and α is the thermal diffusion coefficient. All of these values can be obtained through the simulation in Chemkin-pro software.

As shown in Fig. 9, the effective Lewis number increases with the equivalence ratio, indicating greater flame stability under the rich conditions. In the lean conditions, the effective Lewis number decreases as the hydrogen volume fraction increases, with values falling below 1.0. This suggests that the flame becomes increasingly susceptible to thermal diffusion instability as the Lewis number decreases. Conversely, under rich conditions, the effective Lewis number slightly increases with a

higher hydrogen volume fraction, leading to improved flame stability. Furthermore, a comparison of Fig. 9(a) and Fig. 9(b) shows that, for the same volumetric hydrogen ratio and equivalence ratio, the effective Lewis numbers at initial pressures of 1.5 MPa and 2.5 MPa are nearly identical. This indicates that the initial pressure has minimal impact on diffusional-thermal instability.

Regarding the flame thickness, as illustrated in Fig. 10, the flame thickness initially decreases and then increases with an increase in the equivalence ratio, reaching a minimum value at approximately 1.1. At low volumetric hydrogen ratios, the flame thickness at an equivalence ratio of 0.6 can exceed that at an equivalence ratio of 1.1 by more than a factor of 1000. As the volumetric hydrogen ratio increases, the flame thickness gradually diminishes, suggesting that flames are more prone to hydrodynamic instabilities under conditions characterized by a higher volumetric hydrogen ratio. Furthermore, unlike the effective Lewis number, it is evident from Fig. 10(a) and 9(b) that the flame thickness is significantly influenced by the initial pressure. When both the volumetric hydrogen ratio and equivalence ratio are held constant, lower initial pressures correspond to reduced flame thickness, indicating that flames exhibit increased susceptibility to hydrodynamic instability under higher initial pressure conditions.

Regarding the prediction of the laminar burning velocity of the ammonia-hydrogen mixed fuel, the currently available mechanisms do not encompass all conditions. At low volumetric hydrogen ratios, these mechanisms tend to overestimate laminar burning velocities under rich conditions. Moreover, when the volumetric hydrogen ratio reaches 0.5 or higher, there is a notable absence of suitable mechanisms capable of accurately predicting laminar burning velocities. Hence, a considerable amount of work is required to develop new mechanisms. Additionally, under lean conditions, at high initial pressures, the cellular instability

emerges very early in the spherical flame, even immediately after ignition, making it challenging to obtain the laminar burning velocity. The related methods to obtain the laminar burning velocity under the relevant conditions still need to be further explored.

3.3. Effects of volumetric hydrogen ratios

Fig. 11(a), 11(b), 11(c), and 11(d) show the laminar burning velocities under different volumetric hydrogen ratios at initial pressures of 0.5 MPa, 1.0 MPa, 1.5 MPa, and 2.0 MPa, respectively. The symbols represent experimental results, while the lines represent simulation results using various mechanisms. In order to avoid the influence of flame instability caused by the increase of initial pressure and volumetric hydrogen ratios, only the laminar burning velocity results at lower hydrogen volume fractions are shown in Fig. 11 for high initial pressures. It can be observed that the laminar burning velocity increases with the volumetric hydrogen ratio, exhibiting a nonlinear relationship. With the increase of volumetric hydrogen ratio, the laminar burning velocity increases more and more obviously. When the volumetric hydrogen ratio is low, specifically less than or equal to 0.3, several mechanisms, such as those proposed by Okafor et al. [51] and Gotama et al. [34], provide predictions that closely match the experimental values within the initial pressure range of 0.5 to 2.0 MPa. However,

when the volumetric hydrogen ratio is increased to 0.4 or above, only the Okafor mechanism predicts values lower than the experimental results at initial pressures of 0.5 MPa and 1.0 MPa, while all other mechanisms tend to overestimate the laminar burning velocity.

To investigate the influence of the increasing hydrogen ratio on the reactions of ammonia during combustion, simulations were conducted based on the mechanism proposed by Okafor et al. [51]. The reaction path of ammonia and its intermediate species was analyzed for stoichiometric combustion at an initial pressure of 1.5 MPa with different volumetric hydrogen ratios, as shown in Fig. 12. The black solid lines outline the main reactants, i.e., ammonia, and related nitrogen-containing intermediate species. The arrowheads indicate the direction of the reaction, and the arrows denote the radicals involved in the reaction, along with the proportion of the amount consumed by the reaction relative to the total consumption. The blue, black, and red fonts represent the volumetric hydrogen ratios of 0.1, 0.3, and 0.5, respectively. It can be observed that, regardless of the volumetric hydrogen ratio, the proportion of ammonia and intermediate species such as NH_2 and NH consumed by the reaction is not significantly changed. This indicates that the amount of hydrogen does not significantly affect the reaction path of ammonia itself.

Additionally, it can be seen that the active species OH, H, and O play a significant role in the consumption reactions of ammonia-related

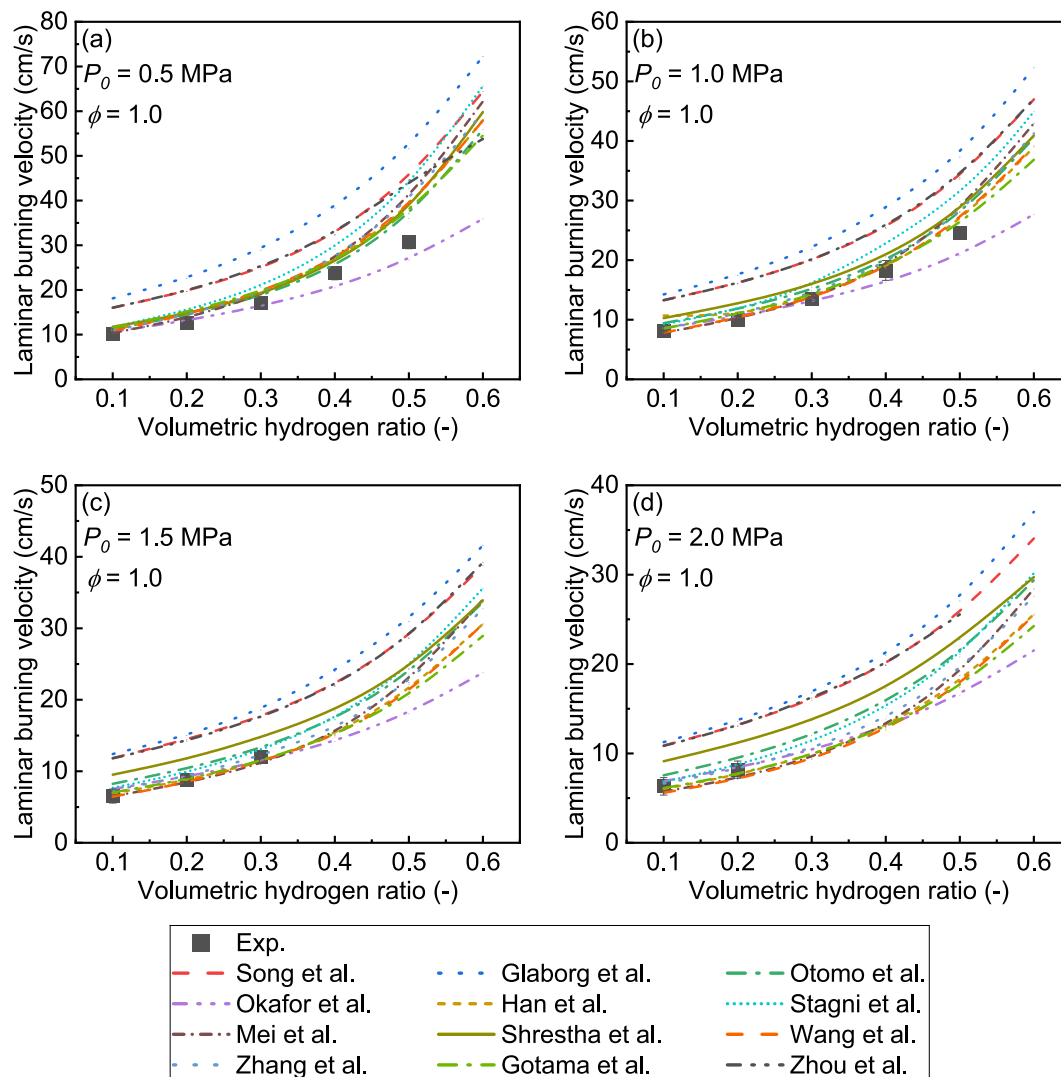


Fig. 11. Laminar burning velocities under different volumetric hydrogen ratios at P_0 of (a) 0.5 MPa, (b) 1.0 MPa, (c) 1.5 MPa and (d) 2.0 MPa. The symbols indicate the experimental results and the lines are the numerical results. ($\phi = 1.0$; $T_0 = 423$ K).

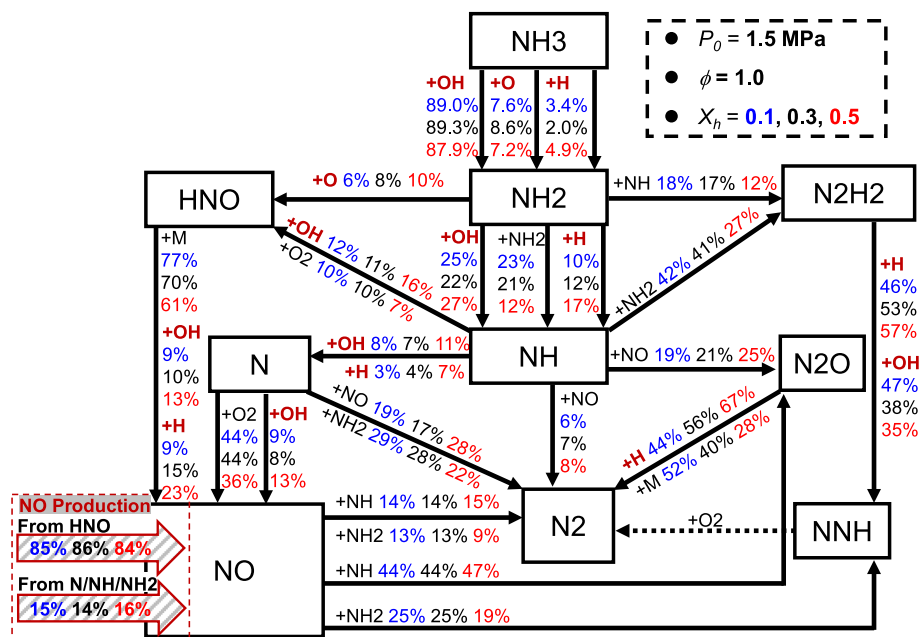


Fig. 12. Reaction pathway of ammonia and the relevant intermediate radicals and the rate of production analysis. ($P_0 = 1.5 \text{ MPa}$; $\phi = 1.0$; $X_h = 0.1, 0.3, 0.5$; $T_0 = 423 \text{ K}$).

substances, especially OH. These species are deeply involved in the reactions of NH_3 and its intermediate groups such as NH_2 , NH , and N_2H_2 . For example, NH_3 primarily undergoes dehydrogenation reactions with OH, O, and H to generate NH_2 , as shown in reactions R1, R2, and R3. NH_2 , NH , and N_2H_2 also undergo similar reactions, as shown in reactions R4 to R6, while N can be oxidized by OH to generate NO, as shown in R7. Thus, it can be seen from the proportion of different substances participating in the reaction in Fig. 12 that the radicals OH, H, and O (especially OH) play a crucial role in promoting the further reactions of NH_3 and its intermediate products.

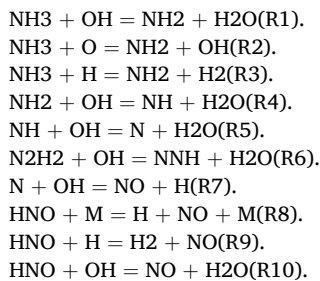


Fig. 13 presents a statistical analysis of the generation rates of OH, O, H, and NO radicals in the ammonia-hydrogen premixed combustion simulation across various volumetric hydrogen ratios, using the same model with the flame speed module in Chemkin-pro. The values depicted in the figure are normalized by their minimum values. As the volumetric hydrogen ratio increases, there is a marked increase in the generation rates of these radicals. Specifically, at a volumetric hydrogen ratio of 0.3, the generation rates for OH, O, and H radicals are approximately double those at a volumetric hydrogen ratio of 0.1, and further increasing the hydrogen fraction to 0.5 results in an approximately fivefold increase compared to 0.1. Consequently, due to its high reactivity, the addition of hydrogen significantly enhances the generation rates of OH, O, and H radicals, thereby increasing their concentration within the active radicals pool. Thus, incorporating hydrogen not only accelerates combustion by substituting the low-reactivity ammonia with the high-reactivity hydrogen but also promotes ammonia's reaction through increased concentrations of OH, O, and H radicals, which play crucial roles in facilitating reactions involving ammonia and its N-containing intermediates.

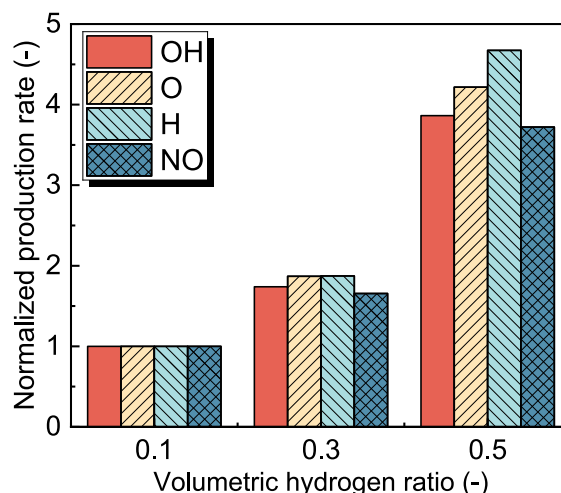


Fig. 13. Normalized production rate of OH, O, H, and NO radicals. ($P_0 = 1.5 \text{ MPa}$; $\phi = 1.0$; $X_h = 0.1, 0.3, 0.5$; $T_0 = 423 \text{ K}$).

For NO, as shown in Fig. 12, NO mainly comes from HNO, N, and NH, of which about 85 % comes from HNO. As Fig. 13 shows, with the increase in hydrogen ratio, the proportion of NO generated and consumed through each elementary reaction remains almost unchanged, which indicates that the introduction of hydrogen does not significantly affect the NO reaction path. As shown in reactions R8, R9, and R10, NO is mainly generated by HNO reacting with H, OH, and M. HNO mainly comes from the oxidation of NH_2 and NH . These reactions are related to radicals such as OH, O, and H, and with the introduction of hydrogen, the amount of these radicals increases significantly, accelerating the NO generation rate. As shown in Fig. 13, when the hydrogen ratio is increased to 0.5, the NO generation rate is increased to 3.7 times that of the hydrogen ratio of 0.1. It should be noted that in the simulation, the mass flow rate supplied at the entrance of the burner was kept constant in this study, and the ammonia mass flow rate decreased by only 9 % as the hydrogen ratio increased from 0.1 to 0.5. As is known, the type of NO in ammonia-related fuels is fuel-type NO, and N comes from the fuel, i.e.,

NH₃. With the introduction of hydrogen, although the amount of ammonia in the reactants is slightly reduced, the NO generation rate still increases significantly. This further indicates that the chemical effects brought about by the introduction of hydrogen have a significant impact on the NO-related reaction rate to promote the NO generation.

3.4. Effects of initial pressure

Fig. 14 illustrates the laminar burning velocities of ammonia-hydrogen mixtures under various initial pressures at a volumetric hydrogen ratio of 0.3. The experimental and simulated results for equivalence ratios of 0.8, 1.0, and 1.2 are presented in Fig. 14(a), 14(b), and 14(c), respectively. It is evident that an increase in the initial pressure leads to a reduction in laminar burning velocity and a nonlinear relationship exists between the laminar burning velocity and initial pressure, with the rate of decrease diminishing at higher pressures. Due to instability issues associated with flames under lean conditions at elevated initial pressures, only results from lower initial pressures are shown for an equivalence ratio of 0.8. No single mechanism accurately predicts the laminar burning velocity across the entire range of initial pressures from 0.1 to 2.5 MPa for these equivalence ratios. At an initial pressure of 0.1 MPa and an equivalence ratio of 0.8, the predictions using the mechanisms by Otomo et al. [50], Song et al. [48], Stagni et al. [53], and Zhou et al. [56] closely align with the experimental results. For the equivalence ratios of 1.0 and 1.2, many mechanisms, including those by Song et al. [48], Otomo et al. [50], Han et al. [52], Stagni et al. [53], Mei et al. [32], Shrestha et al. [21], Wang et al. [54], Zhang et al. [55], and Zhou et al. [56], produce the predicted values very close to the experimental results, while the predictions by Okafor et al. [51] and

Gotama et al. [34] are lower. As the initial pressure increases, the mechanisms by Han et al. [52], Mei et al. [32], Wang et al. [54], Zhang et al. [55], and Okafor et al. [51] provide the results close to the experimental values at an equivalence ratio of 0.8. For the equivalence ratios of 1.0 and 1.2, Okafor's mechanism [51] shows the best prediction performance across the range of initial pressures from 0.5 to 2.5 MPa. The mechanisms by Mei et al. [32], Gotama et al. [34], Wang et al. [54], Han et al. [52], and Zhang et al. [55] perform better at pressures above 1.0 MPa but tend to overestimate at pressures of 0.5 and 1.0 MPa.

The relationship between the laminar burning velocity and the initial pressure is typically represented by Eq. (10) [59],

$$S_L = S_{L,0} \cdot \left(\frac{P_u}{P_{u,0}} \right)^{\beta_p} \quad (10)$$

where $S_{L,0}$ is the laminar burning velocity at the reference initial pressure state ($P_{u,0}$), i.e., $P_{u,0}$ of 0.1 MPa.

Therefore, the pressure-dependent coefficient (β_p) can be obtained by,

$$\beta_p = \ln\left(\frac{S_L}{S_{L,0}}\right) / \ln\left(\frac{P_u}{P_{u,0}}\right) \quad (11)$$

According to the error propagation formula [60], the uncertainty of the pressure-dependent coefficient, $\Delta\beta_p$, can be calculated by,

$$\Delta\beta_p = \left\{ \sum_{i=1}^m \left[\left(\ln \frac{P_{u,i}}{P_{u,0}} - \ln \frac{P_{u,i}}{P_{u,0}} \right) \cdot \frac{\Delta S_{L,i}}{S_{L,i}} \right]^2 \right\}^{0.5} / \left\{ \sum_i \ln^2 \frac{P_{u,i}}{P_{u,0}} - m \left(\ln \frac{P_{u,i}}{P_{u,0}} \right)^2 \right\} \quad (12)$$

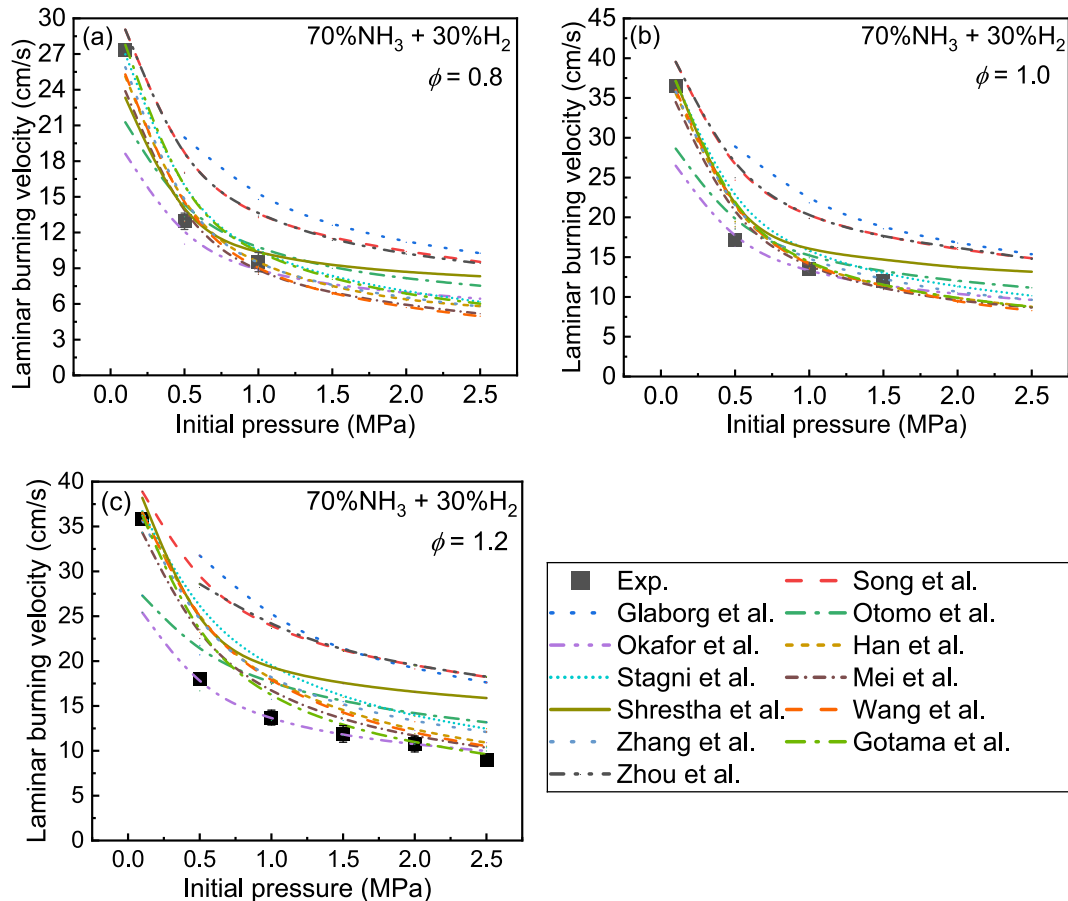


Fig. 14. Laminar burning velocities under different initial pressure at ϕ of (a) 0.8, (b) 1.0 and (c) 1.2. The symbols indicate the experimental results and the lines are the numerical results. ($X_h = 0.3$; $T_0 = 423$ K).

where m is the number of initial pressure state points for the experiment, and $\Delta S_{L,i}$ is the uncertainty of the laminar burning velocity measurement at the initial pressure ($P_{u,i}$).

Fig. 15 shows the experimental results of the pressure dependency of the laminar burning velocity. Symbols represent the experimental data, while the lines denote the fitting results based on these values. The fitting curves exhibit good linearity, with R^2 values all greater than 93 % (with a minimum value of 93.6 %), indicating that the pressure-dependent coefficients are consistent across different initial pressures. Furthermore, by comparing the slopes of the curves, it is evident that the pressure-dependent coefficients, $|\beta_p|$, are very similar for different equivalence ratios. Fig. 16 further illustrates how the pressure-dependent coefficient varies with the equivalence ratio. It indicates that as the equivalence ratio increases from lean to rich conditions, the pressure-dependent coefficient first increases and then decreases, reaching a maximum at an equivalence ratio of 1.1. However, the overall numerical fluctuations are relatively small, ranging between -0.5 and -0.4 . The negative value signifies that the laminar burning velocity decreases as the initial pressure increases. The pressure-dependent coefficient is a crucial parameter in many combustion simulation settings, affecting the accuracy of combustion predictions under varying pressure conditions. This data provides valuable support for simulating combustors across a wide range of pressure conditions, covering the conditions under high pressure.

Fig. 17 shows the sensitivity coefficients of the laminar burning velocity at a volumetric hydrogen ratio of 0.3 under different initial pressures. The numerical values are normalized using the maximum value. In the range of 0.1 to 2.5 MPa, the elementary reaction, $H + O_2 = O + OH$, as a key chain-branching reaction, has the largest positive sensitivity coefficient. However, as the initial pressure increases, the sensitivity coefficient of this reaction gradually decreases. Reactions involving OH and H, such as $H + NO + M = HNO + M$, $NH_2 + NO = NNH + OH$, and $OH + H_2 = H + H_2O$, also exhibit positive sensitivity coefficients, with the sensitivity coefficients of these reactions changing differently as the initial pressure increases. In the combustion of ammonia, NH_2 is a key intermediate product. The dehydrogenation reaction $NH_3 + M = NH_2 + H + M$ is a three-body pressure-dependent reaction with the largest negative sensitivity coefficient, and its sensitivity increases with the pressure. Additionally, the chain termination reaction $HNO + H = H_2 + NO$ and the chain branching reaction $OH + HO_2 = H_2O + O_2$ also have negative sensitivity coefficients, but their

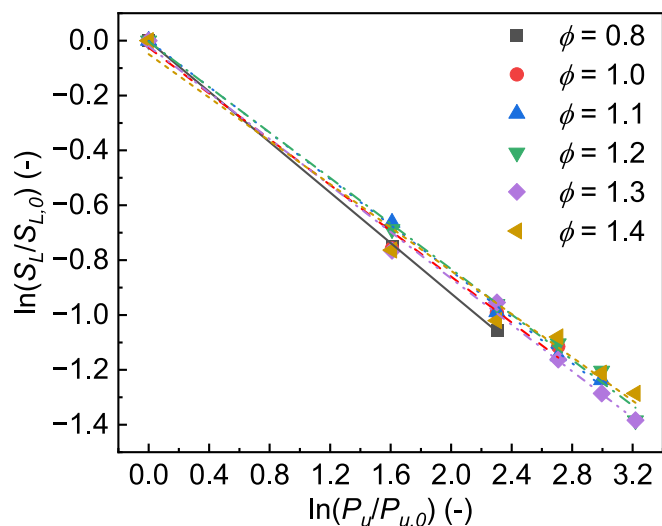


Fig. 15. Experimental results of pressure dependence of laminar flame propagation for different equivalence ratios. The symbols represent the experimental results, while the lines represent the result of fitting the experimental values. ($X_h = 0.3$; $T_o = 423$ K).

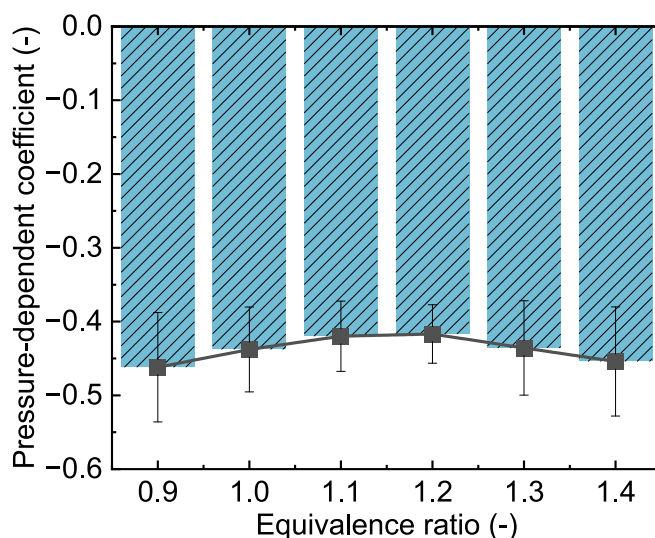


Fig. 16. Pressure-dependent coefficients under different equivalence ratios. ($X_h = 0.3$; $T_o = 423$ K).

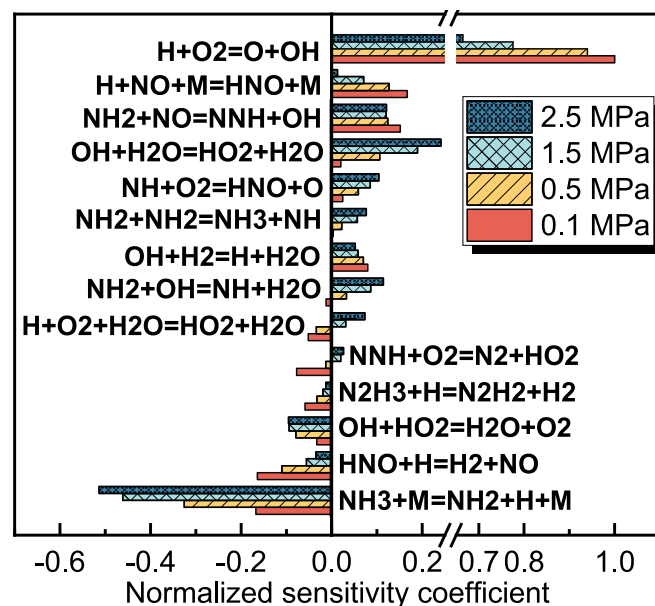


Fig. 17. Sensitivity analysis of Laminar burning velocities under the P_o of 0.1, 0.5, 1.5 and 2.5 MPa. ($\phi = 1.0$; $X_h = 0.3$; $T_o = 423$ K).

sensitivity to laminar burning velocity is relatively weak. It is noteworthy that reactions such as $NH_2 + OH = NH + H_2O$, $H + O_2 + H_2O = HO_2 + H_2O$, and $NNH + O_2 = N_2 + HO_2$ exhibit negative sensitivity coefficients at low initial pressures but transition to positive sensitivity coefficients as the initial pressure increases. This indicates that these reactions have opposite characteristics under different initial pressures. To develop a mechanism applicable across a range of initial pressures, from low to high, it is essential to carefully consider the relevant data for the reactions illustrated in Fig. 17.

4. Summary and conclusions

The laminar burning characteristics under different equivalence ratios, volumetric hydrogen ratios, and initial pressures were investigated through the outwardly propagating spherical flame experiments to expand the database of laminar burning velocities for ammonia and hydrogen mixtures at high pressures up to 2.5 MPa. The experimental

results for laminar burning velocities across a broad range of operating conditions were compared with simulation results obtained using several advanced mechanisms to assess the applicability of these mechanisms. The chemical reaction mechanisms of ammonia-hydrogen combustion at elevated pressures were investigated, leading to a clearer understanding of the associated chemical kinetics. The major findings are summarized as follows.

- The laminar burning velocity initially increases and then decreases as the equivalence ratio rises, reaching its maximum value around an equivalence ratio of 1.1 for all specified volumetric hydrogen ratios and initial pressures.
- The laminar burning velocity shows a nonlinear increase with the volumetric hydrogen ratio. As the volumetric hydrogen ratio rises, the rate of increase in laminar burning velocity becomes more pronounced. Analysis of the reaction pathway for the ammonia-hydrogen combustion indicates that introducing hydrogen enhances ammonia combustion by increasing the concentration of active radicals, such as OH, O, and H in the reactive pool, thereby accelerating the overall combustion process of ammonia.
- The laminar burning velocities of ammonia-hydrogen mixtures were determined at high initial pressures up to 2.5 MPa, significantly expanding the database of laminar burning velocities for these mixtures. The laminar burning velocity exhibits a nonlinear decrease with the increasing initial pressure, with the rate of reduction becoming progressively slower as the initial pressure increase. Additionally, the pressure-dependent coefficient of the laminar burning velocity was derived, providing a key reference for establishing practical combustion models, which is typically a critical parameter required in combustion simulations.
- The thermal diffusion instability is more prevalent under lean conditions, attributed to the reduced effective Lewis number. At high volumetric hydrogen ratios and elevated initial pressures, the flame's susceptibility to hydrodynamic instabilities increases due to its diminished thickness.
- The predictions derived from various mechanisms exhibit notable variability. The mechanisms proposed by Okafor et al. and Gotama et al. demonstrate reasonable accuracy under specific conditions, particularly at volumetric hydrogen ratios of 0.3 and 0.4, while discrepancies remain at other volumetric hydrogen ratios, especially under low-pressure conditions.
- The sensitivity analysis of the laminar burning velocity elucidates the critical elementary reactions that are sensitive to varying initial pressures, thereby providing robust data to support the optimization of ammonia-hydrogen chemical kinetics models and the development of simulation models for practical combustors.
- The current mechanisms are insufficient for making accurate predictions across all conditions presented in this study. Specifically, at low volumetric hydrogen ratios, these mechanisms tend to overestimate the laminar burning velocities at rich conditions. Furthermore, when the volumetric hydrogen ratios reaches 0.5 or higher, there is a lack of suitable mechanisms capable of accurately predicting laminar burning velocities. Referring to the results of this study to develop an advanced mechanism applicable to a wide range of conditions for ammonia-hydrogen combustion is essential. Through reaction path analysis and sensitivity analysis, this study have delineated a pathway for the development of future mechanisms.

CRediT authorship contribution statement

Ning Wang: Writing – original draft, Methodology, Investigation, Data curation, Conceptualization. **Tie Li:** Writing – review & editing, Supervision. **Xinpeng Guo:** Methodology, Investigation. **Zehao Wu:** Methodology, Investigation. **Shuai Huang:** Methodology, Investigation. **Xinyi Zhou:** Methodology, Investigation. **Shiyan Li:** Methodology. **Run**

Chen: Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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